

Product Requirements Document

MVP — v1.0

B2B · MANUFACTURING SME

VERSION:	1.0
DATE:	April 23, 2026
STATUS:	Draft
SEGMENT:	Manufacturing SMEs (10–500 employees)
PLATFORM:	Web App + Mobile PWA
MODEL:	Per-Seat SaaS + Enterprise

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SECTION 1

Executive Summary

Problem Statement

Manufacturing SMEs (10–500 employees) are losing money to unplanned equipment downtime, lost tools, and missed maintenance windows. Most track assets via spreadsheets or paper-based systems. Enterprise solutions like IBM Maximo and SAP are too expensive, too complex, and too slow to implement for this segment.

Solution

AssetFlow is a SaaS asset management platform built specifically for manufacturing SMEs. It combines asset inventory tracking and preventive maintenance scheduling in a single, easy-to-use web and mobile PWA — no installation required, no IT team needed.

Market Opportunity

Metric	Data
Global market size by 2028	\$32.2 billion (CAGR 10.1% — Grand View Research)
Highest demand vertical	Manufacturing & Industrial
Primary gap	Existing tools are too expensive (enterprise) or too generic (SME)
Target addressable market	~2.5M manufacturing SMEs in North America + Europe
BFSI market share (2025)	22.81% revenue share — largest single segment
Healthcare growth	7.88% CAGR through 2031 — best expansion target for v2

Business Model Summary

- Per-seat SaaS subscription — Starter and Pro tiers billed monthly or annually
- Enterprise plan sold via direct sales with custom pricing and dedicated support
- Revenue model: Monthly Recurring Revenue (MRR) with annual contract option for enterprise

SECTION 2

Product Vision & Goals

Vision Statement

"Give every manufacturing team the visibility and control over their physical assets that only large enterprises could afford before — at a price any SME can justify."

MVP Objectives

- Allow any manufacturing SME to register, tag, and track all physical assets in under one hour
- Enable technicians to scan assets in the field and log maintenance work from their phone
- Give managers a live dashboard to monitor asset health and upcoming maintenance
- Automate maintenance reminders to prevent unplanned downtime

Success Metrics — 6 Months Post-Launch

Metric	Target
Monthly Active Customers	50 paying businesses
Average Seats per Customer	8 users
Monthly Recurring Revenue	\$25,000 MRR
Churn Rate	< 5% monthly
Asset Scan Actions / Day	500+ across all customers
Maintenance Tasks Completed On-Time	> 80%
NPS Score	> 40

Out of Scope — MVP

- ❌ Third-party integrations (QuickBooks, Slack, SAP, REST API)
- ❌ Native iOS or Android app (PWA covers mobile for MVP)
- ❌ B2C individual tracking features
- ❌ AI-powered predictive maintenance
- ❌ Multi-site enterprise hierarchy beyond basic org management
- ❌ Offline mode (requires connectivity for MVP)

SECTION 3

Market Context

Target Customer

Primary segment	Manufacturing and industrial SMEs with physical equipment, machinery, vehicles, or tools
Company size	10–500 employees
Geography	North America (primary), English-speaking markets (secondary)
Budget authority	Operations Manager, Plant Manager, or Business Owner
Current state	Managing assets in spreadsheets, paper logs, or nothing at all

Competitive Landscape

Competitor	Strength	Weakness	AssetFlow Advantage
IBM Maximo	Feature-rich, enterprise-grade	\$100K+ implementation, 6-month setup	10x cheaper, same-day setup
UpKeep	Mobile-first, easy to use	Maintenance-only, weak asset registry	Unified registry + maintenance
Fiix	Good maintenance scheduling	Complex UI, steep learning curve	Simpler UX, faster onboarding
Limble CMMS	Great for maintenance teams	No GPS, limited asset tracking	GPS tracking + full asset registry
Spreadsheets	Free, familiar	No automation, error-prone, no mobile	Automated alerts + mobile scanning

Positioning Statement

AssetFlow is the only asset management tool built for manufacturing SMEs that combines physical asset tracking (registry, QR/barcode scanning, GPS) with preventive maintenance scheduling in a single platform — accessible from desktop and mobile without an app store install.

SECTION 4

User Personas

AssetFlow serves three distinct roles within a manufacturing organisation. Each persona has different goals, frustrations, and daily workflows.

■ Admin — Operations Manager

Sarah, 42 · 80-person metal fabrication company

Wants a single source of truth for all equipment — what they own, where it is, and when it needs servicing. Currently drowning in spreadsheets that go out of date the moment they are shared.

Key jobs to be done:

- Register all company assets once (bulk import or one-by-one)
- Assign assets to locations, departments, or individuals
- Configure maintenance schedules for each asset type
- Manage user accounts and permissions
- View audit history for compliance

■ Technician — Field Worker

Marcus, 28 · Equipment Technician

Needs to know what maintenance tasks are due today and log completion quickly — without paperwork. Currently gets paper work orders that get lost and has no easy way to find asset history on the floor.

Key jobs to be done:

- Scan a QR code on a machine and immediately see its history and next service date
- Log a completed maintenance task in under 60 seconds
- Report an issue or flag an asset as out of service
- View assigned work orders for the day

■ Manager — Plant Manager

David, 55 · Plant Manager

Wants to prevent downtime and know at a glance which assets are healthy, overdue, or flagged. Only finds out about equipment problems after a breakdown. Manual reports from spreadsheets take hours.

Key jobs to be done:

- View a dashboard showing asset health status across the facility
- See upcoming and overdue maintenance tasks
- Approve work orders and assign them to technicians
- Export a monthly maintenance report for the board

SECTION 5

User Journeys

5.1 Admin Journey — Onboarding & Setup

1. Sign Up

- Creates account, sets up company profile (name, industry, location)

2. Add Assets

- Imports assets via CSV upload or adds individually
- Assigns each asset: name, category, serial number, location, purchase date

3. Generate QR / Barcode Labels

- System generates printable QR codes for each asset
- Admin prints and physically attaches labels to equipment

4. Configure Maintenance Schedules

- Sets recurring maintenance tasks per asset or asset category
- Defines interval (daily, weekly, monthly, by usage hours)
- Assigns default technician responsible

5. Invite Team

- Sends email invites to Technicians and Managers
- Assigns roles and permissions per user

6. Verify Setup

- Reviews asset list, scans a test QR code, confirms alerts are working

5.2 Technician Journey — Daily Work

1. Start of Shift

- Opens AssetFlow on mobile browser (PWA)
- Views 'My Work Orders' — tasks assigned for today

2. Scan an Asset

- Taps 'Scan Asset' — camera opens
- Scans QR code on machine
- Sees asset profile: name, location, last service date, status, open issues

3. Log Maintenance

- Taps 'Log Maintenance' on the asset
- Selects task type (from predefined list or free text)
- Adds notes, photos (optional), marks duration
- Submits — work order auto-closes if linked

4. Report an Issue

- Taps 'Report Issue' on scanned asset
- Selects severity (Low / Medium / Critical)
- Adds description and photo
- Submits — manager gets alert immediately for Critical issues

5. End of Shift

- Views completion status of assigned tasks
- Any incomplete tasks automatically flagged for manager

5.3 Manager Journey — Monitoring & Oversight

1. Morning Dashboard Review

- Opens AssetFlow on desktop browser
- Sees summary: total assets, assets due for maintenance, overdue count, open issues
- Red/amber/green status indicators per asset category

2. Review Upcoming Maintenance

- Views maintenance calendar — next 7 / 30 days
- Sees which tasks are assigned, unassigned, or overdue
- Reassigns or approves work orders as needed

3. Respond to Alerts

- Receives email/in-app alert for critical issues or overdue tasks
- Reviews issue details, adds notes, changes priority
- Assigns or escalates to another technician

4. Monthly Reporting

- Navigates to Reports section
- Views maintenance completion rate, asset downtime events, work order history
- Exports PDF or on-screen summary for board/management review

SECTION 6

Feature Specifications

6.1 Asset Registry & Management

Central database of all company assets with full profile per asset.

- Each asset record contains: name, asset ID (auto-generated), category, serial number, manufacturer, model, purchase date, purchase cost, warranty expiry, location, assigned user, status, and custom fields (Admin-configurable key-value pairs, up to 10 per asset)
- Bulk import via CSV with field mapping wizard
- Asset categories are user-configurable (e.g., Machinery, Vehicles, Tools, IT Equipment)
- Assets can be linked to a physical location (building, floor, zone) defined by Admin
- Full edit history per asset (who changed what and when)
- Asset duplication for creating similar assets quickly
- Archive/retire assets without deletion (preserves history)
- Search and filter by any field; sortable columns in list view

6.2 QR / Barcode Scanning

Camera-based scanning on mobile PWA to identify assets instantly in the field.

- Admin generates a unique QR code per asset — printable, downloadable as PNG or PDF sheet
- Technician taps 'Scan' in mobile browser — uses device camera (no app install needed)
- Scanning opens the asset profile immediately
- System also accepts manual asset ID entry as fallback
- QR codes are permanent and tied to the asset ID (survive asset edits)
- Batch QR label printing: print labels for multiple assets at once (Avery-compatible sheet layout)
- Compatible with standard USB barcode scanners for fixed workstations

6.3 GPS / Location Tracking

Track the physical location of mobile assets using browser GPS. Available on Pro and Enterprise plans only.

- When a Technician scans or checks in an asset from their mobile device, GPS coordinates are optionally recorded
- Admin can view last known location of each asset on an embedded map view (OpenStreetMap / Leaflet.js)
- Location history log per asset (timestamp + coordinates)
- GPS is opt-in: Technician is prompted for location permission on first use
- Assets can have a fixed 'home location' defined by Admin for reference
- Location capture only occurs on user action (scan or check-in), not continuous background tracking

6.4 Maintenance Scheduling

Define recurring preventive maintenance tasks per asset and track completion.

- Admin creates maintenance schedules per asset or asset category

- Schedule types: time-based (every N days/weeks/months) or usage-based (every N hours — Technician manually logs current hours when completing a work order; Admin sets the threshold)
 - Each schedule defines: task name, description, estimated duration, assigned technician (default), and priority
 - System auto-generates work orders based on schedule — N days before due date (configurable lead time)
 - Maintenance calendar view: month/week/day views showing upcoming and overdue tasks
 - Maintenance history per asset: complete log of all past tasks with timestamps and technician notes
 - Bulk schedule assignment: apply a maintenance template to an entire asset category
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6.5 Work Order Management

Task-level tracking for maintenance and repair jobs from creation to completion.

- Work orders can be auto-generated (from maintenance schedule) or manually created by Admin/Manager
 - Work order fields: title, asset linked, assigned technician, due date, priority (Low/Medium/High/Critical), description, estimated hours, status
 - Status flow: Open → In Progress → Completed / Cancelled
 - Technician updates status and adds completion notes from mobile
 - Photo attachments per work order (up to 5 photos, max 10MB each)
 - Manager can add comments or reassign at any stage
 - Overdue work orders automatically escalate (change to red status + notification)
 - Work order history retained indefinitely per asset
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6.6 Automated Alerts & Notifications

Proactive notifications to prevent missed maintenance and surface critical issues.

- In-app notifications (bell icon + notification feed) and email notifications (HTML formatted, configurable per user)
 - Triggers: Maintenance due in N days · Work order overdue · Critical issue reported · Asset status changed · Work order assigned to technician
 - Users configure their own notification preferences (which events, in-app vs email)
 - Admin can set org-level defaults for notification rules
 - No SMS or push notifications in MVP (email + in-app only)
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6.7 Dashboard & Reporting

Real-time visibility into asset health and maintenance performance for Managers.

- Dashboard widgets: Total assets by status · Maintenance due this week/month · Overdue maintenance count · Open work orders by priority · Recent activity feed · Assets with open critical issues
 - Reports: Maintenance completion rate · Asset downtime log · Work order history (filterable) · Full asset inventory export
 - All reports exportable as PDF or CSV
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6.8 User Management & Roles

Role-based access control with three permission levels.

- Admin invites users via email; invited users set their own password
- Each seat = one active user account
- Admins can deactivate users (frees up the seat without deleting history)
- Each company account is isolated — no cross-company data access
- Three roles: Admin (full access) · Manager (operational) · Technician (field tasks only)

SECTION 7

Role × Feature Matrix

Feature	Admin	Manager	Technician
Asset Registry	Full access	Read only	Read only (via scan)
QR Code Generation	■	■	■
GPS Location View	■	■	Logs own location
Maintenance Scheduling	■	Create & edit	View assigned
Work Orders	Full CRUD	Create & assign	Update & complete
Issue Reporting	■	■	■
Manager Dashboard	■	■	■
Reports & Exports	■	■	■
User Management	■	■	■
Notification Config	Org-wide	Own settings	Own settings
Billing & Subscription	■	■	■

SECTION 8

Technical Requirements

Platform

Requirement	Specification
Deployment	SaaS, cloud-hosted
Web App	Responsive web application (desktop-first, mobile-compatible)
Mobile	Progressive Web App (PWA) — installable from browser, no app store
Offline support	None (MVP requires connectivity)
Browser support	Chrome, Firefox, Safari, Edge (latest 2 major versions)
QR Scanning	Browser camera API (getUserMedia) — no native app required
GPS	Browser Geolocation API (permission-based, user-triggered)
Map display	Embedded map via OpenStreetMap / Leaflet.js
Authentication	Email + password with secure session management; MFA optional
Data storage	Cloud database, multi-tenant with strict tenant isolation
File storage	Cloud object storage for photos and QR label PDFs

Performance Targets

Metric	Target
Page load time (web)	< 2 seconds (P90)
QR scan to asset profile	< 1 second
Dashboard load	< 3 seconds
API response time	< 500ms (P95)
Uptime SLA	99.5% monthly

Security Requirements

- All data encrypted in transit (TLS 1.2+) and at rest (AES-256)
- Tenant data strictly isolated — no cross-account data leakage possible
- Role-based access enforced server-side (not just at the UI layer)
- Password requirements: minimum 8 characters, enforced on create/reset
- Session timeout after 24 hours of inactivity

- Audit log for all data changes (who, what, when) — retained 12 months

Scalability

Constraint	Limit
Concurrent users	Up to 10,000 at MVP launch
Assets per tenant	Up to 100,000 asset records
Storage per tenant	Up to 50GB for photo attachments

SECTION 9

Pricing & Business Model

Subscription Tiers

Feature	Starter \$15/user/mo	Pro \$28/user/mo	Enterprise Custom
Minimum seats	3	5	20
Asset limit	200	Unlimited	Unlimited
User limit	10	Unlimited	Unlimited
QR / Barcode Scanning	■	■	■
GPS Tracking	■	■	■
Maintenance Scheduling	■	■	■
Work Orders	■	■	■
Reporting & Export	Basic	Full	Full + Custom
Support	Email	Priority email	Dedicated CSM
SLA	99.5%	99.5%	99.9%
Onboarding	Self-serve	Self-serve	White-glove setup
Contract	Monthly	Monthly / Annual	Annual

Enterprise Plan Details

- Sold via direct sales (outbound + inbound from website)
- Custom pricing based on seat count, asset volume, and contract length
- Includes: dedicated Customer Success Manager, custom onboarding and data migration, priority support with 4-hour SLA
- Annual contract with Net-30 invoicing and volume discounts for 50+ seats

Revenue Projections — Month 6

Tier	Customers	Avg Seats	MRR
Starter	30	5	\$2,250
Pro	15	10	\$4,200
Enterprise	3	25	\$7,500+
Total	48	—	~\$14,000–\$25,000

SECTION 10

MVP Timeline & Milestones

6-Month Build Plan

Phase	Duration	Deliverables
Phase 1: Foundation	Weeks 1–4	Auth system, company onboarding, asset registry (CRUD), basic user management
Phase 2: Scanning & Tracking	Weeks 5–8	QR code generation, camera-based PWA scanning, GPS location capture, map view
Phase 3: Maintenance Core	Weeks 9–12	Maintenance scheduling engine, work order creation & assignment, status flows
Phase 4: Alerts & Dashboard	Weeks 13–16	Email + in-app notifications, manager dashboard, alert rules
Phase 5: Reports & Polish	Weeks 17–20	Reporting module, PDF/CSV export, mobile PWA polish, performance tuning
Phase 6: Launch Prep	Weeks 21–24	Security audit, billing integration (Stripe), pricing page, onboarding flow, beta testing with 3–5 design partners

Key Milestones

- ◆ **Week 4:** Internal demo of core asset registry — validate with 2 potential customers
- ◆ **Week 8:** First QR scan working end-to-end on mobile
- ◆ **Week 12:** First complete maintenance work order lifecycle demo
- ◆ **Week 20:** Feature-complete beta ready for design partners
- ◆ **Week 24:** Public launch — Starter and Pro plans live

Go-To-Market

- Launch on Product Hunt + relevant manufacturing communities (r/manufacturing, r/lean)
- Outbound to manufacturing SMEs via LinkedIn Sales Navigator
- Content marketing: 'The spreadsheet-to-AssetFlow migration guide'
- Design partner program: 5 early customers get 6 months free in exchange for feedback and case studies
- Enterprise leads handled by founder directly at launch

Document prepared for internal product planning and investor review. All projections are estimates based on market research and comparable SaaS benchmarks.